

Change Log: Cafe Sales Data Cleaning

DecodeLabs Data Analytics | Project 1: Data Cleaning & Preparation | Tool: Microsoft Excel (formula-driven workbook)
Dataset: Cafe Sales - Dirty Data for Cleaning Training (Kaggle). Prepared 9 October 2026.

If it isn't documented, it didn't happen: every change below shows what was changed, how many rows it touched, and why.

1. Summary

Measure	Result
Rows in raw dataset	10,000
Rows in cleaned dataset	10,000 (no rows deleted; no duplicate IDs found)
Cleaned file	cafe_sales_clean.csv (also the Final_Clean sheet of the workbook)

2. Verification gate (required for Project 2)

Check	Before	After	Status
Duplicate Transaction IDs (extra copies)	0	0	PASS
Incorrectly formatted dates (not YYYY-MM-DD)	0	0	PASS
Rows where Total Spent does not equal Quantity x Price	0	0	PASS
Non-positive Quantity or Price values	0	0	PASS
Items not in the Lookups item list	0	0	PASS
Items whose price varies (reference price unreliable)	0	n/a	PASS
Rows with missing Transaction ID	0	0	PASS
Rows with missing Transaction Date	460	460	REVIEW
Rows still missing Quantity, Price or Total (not derivable)	6	6	REVIEW
Formula cells found in Final_Clean A:H (must be pasted VALUES)	n/a	0	PASS
Row reconciliation: (Raw rows - duplicates) vs Final rows	10000	10000	PASS

Result: 0 duplicate Transaction IDs and 0 incorrectly formatted dates. Independently re-checked on the exported CSV: all 10,000 IDs are unique and match the pattern TXN_ plus 7 digits; all 9,540 populated dates are valid YYYY-MM-DD values (range 2023-01-01 to 2023-12-31). Dates that are blank (REVIEW) were missing in the raw data and are not incorrectly formatted.

3. Change log

Change ID	Description	Rows	Impact	Status
CR001	Trimmed leading/trailing/extra spaces in all columns	0	Prevents 'Cash ' and 'Cash' counting as different values	Resolved
CR002	Converted placeholder values (ERROR, UNKNOWN, N/A...) to missing	3,256	Makes missing data detectable and countable	Resolved
CR003	Identified truly blank cells as missing (see Audit section B)	6,826	Gaps are now documented, not hidden	Resolved
CR004	Removed duplicate Transaction IDs (kept first occurrence)	0	One record per transaction; counts no longer inflated	Resolved
CR005	Standardised Transaction Date to ISO 8601 (YYYY-MM-DD)	9,540	Single, sortable date format	Resolved
CR006	Dates missing in the raw data (ERROR/UNKNOWN placeholders, blanks or unparseable text) left blank, not guessed	460	Dates are never guessed; excluded from time analysis only	Open - needs review

Change ID	Description	Rows	Impact	Status
CR007	Standardised Item, Payment Method, Location to Proper Case	3,017	Merges case variants into one category	Resolved
CR008	Inferred missing Item from its price where that price belongs to exactly one item	489	Preserved 489 records	Resolved
CR009	Filled remaining missing Item with the fill value from Lookups	480	Preserved 480 records	Resolved
CR010	Filled missing Payment Method with the fill value from Lookups	3,178	Preserved 3,178 records	Resolved
CR011	Filled missing Location with the fill value from Lookups	3,961	Preserved 3,961 records	Resolved
CR012	Filled missing Price Per Unit (Total/Quantity, else item reference price)	527	Preserved 527 records	Resolved
CR013	Filled missing Quantity (Total/Price, else median; see CR017 for rows still incomplete)	479	Preserved 479 records	Resolved
CR014	Derived missing Total Spent as Quantity x Price	499	Preserved 499 records	Resolved
CR015	Corrected Total Spent where it differed from Quantity x Price	0	Fixes internally inconsistent revenue figures	Resolved
CR016	Rounded Price and Total to 2 decimals; Quantity to whole numbers	10,000	Consistent numeric precision	Resolved
CR017	Rows where Quantity, Price or Total could not be derived (left blank where unknown)	6	Not guessed; listed for review	Open - needs review

4. Missing values found in the raw data

Column	ERROR	UNKNOWN	Blank	Total missing	% of rows
Transaction ID	0	0	0	0	0.0%
Item	292	344	333	969	9.7%
Quantity	170	171	138	479	4.8%
Price Per Unit	190	164	179	533	5.3%
Total Spent	164	165	173	502	5.0%
Payment Method	306	293	2,579	3,178	31.8%
Location	358	338	3,265	3,961	39.6%
Transaction Date	142	159	159	460	4.6%
Total	1,622	1,634	6,826	10,082	

5. Method decisions and open items

Nothing was deleted. Missing values were filled where they could be derived from the row itself (Total / Quantity, Total / Price, Quantity x Price), then from the item's reference price (every item has exactly one price in this dataset, verified), then with the median quantity (3) as a last resort.

Missing Item was inferred from price only where that price belongs to exactly one item (1.00 Cookie, 1.50 Tea, 2.00 Coffee, 5.00 Salad); prices 3.00 and 4.00 are shared by two items each, so those rows stay 'Unknown'. **Payment Method (31.8% missing) and Location (39.6% missing)** were labelled 'Unknown' rather than filled with the most common value, which would have distorted the category shares.

Open item (CR006): 460 dates were missing in the raw data (301 ERROR/UNKNOWN, 159 blank). They are left blank, not guessed, and are excluded from time-based analysis only.

Open item (CR017): 6 rows cannot have Quantity, Price and Total fully derived because too many of their fields are missing: TXN_3611851, TXN_7524977, TXN_9646000 (Quantity known only) and TXN_7376255, TXN_1082717, TXN_1208561 (Total known, Price unknown; Quantity set to the median of 3 in these three). They are retained and flagged for stakeholder review.